

Lower-Limb Joint Angle Prediction Using a CNN-BiLSTM Model

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Abstract—This project builds upon a previously developed multisensor gait data collection and preprocessing pipeline to design and train a neural network model capable of predicting the hip, knee, and ankle joint angles for one future gait cycle. Using the Vicon motion capture system, electromyography (EMG) sensors, inertial measurement units (IMU), and force plates, gait cycles were segmented using heel-strike events and normalized to a fixed gait cycle percentage scale. The resulting time-series gait data were reshaped to form suitable inputs and outputs for neural network training. To reduce output dimensionality, principal component analysis (PCA) was applied to the knee and ankle joint angles, reducing the three x, y, z components to a single dominant component. The consecutive gait cycles were then reorganized into sequence-based training batches. A hybrid convolutional and bidirectional long short-term memory (CNN-BiLSTM) neural network was designed and trained using the data from 7 participants in the previous term, to predict one future gait cycle from multiple preceding cycles. Across five independent training runs, the model achieved a mean RMSE of 0.045 ± 0.002 rad (approximately 2.58°), indicating accurate and consistent predictive performance. Among the outputs, the knee PCA component exhibited the highest mean RMSE of 0.064 rad (3.67°), while the hip Y component achieved the lowest mean RMSE of 0.018 rad (1.03°). The resulting forecasts of lower-limb joint trajectories from wearable sensor data provide a foundation for future real-time gait prediction and rehabilitation applications.

Index Terms—wearable robotics, exoskeleton, joint angle prediction, neural network, CNN-LSTM

I. INTRODUCTION

As the field of neural networks continues to evolve rapidly, the integration of neural networks with other systems has been increasingly studied and widely researched. Biomechanics is no exception to this integration with artificial intelligence [1]. Neural networks are commonly categorized into two types based on their purpose: classification and regression.

The main objective of this project is to predict lower-limb joint angles (hip, knee, and ankle) for one future gait cycle, given past multi-sensor data. Therefore, this problem is categorized as a time-series regression task, where both temporal dependencies and gait patterns should be captured [2].

A hybrid CNN-biLSTM architecture is commonly used where convolutional neural networks are used for pattern recognition in gait signals, and bidirectional long short-term

memory networks are used to model temporal dependencies within the gait cycle [2].

II. METHODS

A. Model I/O

The goal of this model is to predict joint angle trajectories for one future gait cycle, given previous gait data. For each leg (left or right), the model consists of 35 input feature channels and 5 output channels. The details are listed as follows:

Input: 35 channels

- 5 channels: EMG sensors (Medial Head of Gastrocnemius, Tibialis Anterior, Rectus Femoris, Biceps Femoris Long Head, Gluteus Maximus)
- 18 channels: Leg IMUs (thigh, shank, foot) (3 sensors $\times$ (3-axis acceleration + 3-axis gyroscope))
- 6 channels: Center chest IMU (3-axis acceleration + 3-axis gyroscope)
- 6 channels: Lower back IMU (3-axis acceleration + 3-axis gyroscope)

Output: 5 channels

- 3 channels: Hip joint angles (X, Y, Z axes)
- 1 channel: Knee joint angle (reduced principal component)
- 1 channel: Ankle joint angle (reduced principal component)

B. Principal Component Analysis (PCA)

Hip joint angles are preserved along all three axes, as the hip is a ball-and-socket joint with multi-planar motion. In contrast, the knee and ankle joints exhibit predominantly single-axis rotational behavior. Therefore, Principal Component Analysis (PCA) is applied to extract the dominant rotational component and reduce output dimensionality.

The PCA procedure is performed as follows:

1. Global PCA Fitting

- Stack knee joint angle samples across all files
- Stack ankle joint angle samples across all files
- Fit PCA with $n = 1$ component for each joint (using `scikit-learn`)
- Save the fitted PCA models for knee and ankle

2. PCA Application to Each File

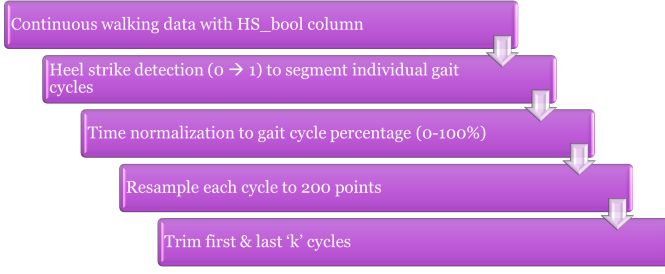


Fig. 1: Gait cycle segmentation steps

- Transform 3-axis joint angles into a 1D PCA representation
- Use the resulting 1D signal as the regression target instead of predicting all three axes

Final model output targets:

$$[\text{Hip}_X, \text{Hip}_Y, \text{Hip}_Z, \text{Knee}_{PCA}, \text{Ankle}_{PCA}]$$

C. Gait Cycle Segmentation

Gait cycles are segmented using force plate data. For each participant file, a new column (HS_bool) was created to indicate heel-strike events. This boolean variable represents whether ground reaction force exceeds a defined threshold.

Heel-strike events are detected by identifying the rising edge of HS_bool, corresponding to the initial heel-contact phase. Consecutive heel-strike events are then used to define individual gait cycles.

Each segmented gait cycle then undergoes time normalization by converting it to gait cycle percentage (0–100%), and each cycle is resampled to 200 points using interpolation.

Finally, the first and last k gait cycles of each trial are removed to reduce transient effects and potential inconsistencies at the beginning and end of the recording.

Figure 1 shows the steps visually.

D. Sliding Window Batch Generation

To construct supervised training samples, a sliding window approach is applied at the gait cycle level. The objective is to use n consecutive gait cycles as the input window and predict the $(n+1)^{\text{th}}$ gait cycle as the output. For this project, $n = 10$. The window is shifted forward by one gait cycle at a time to generate multiple training samples.

The batch generation process is illustrated in Table I, where N represents the last valid gait cycle.

TABLE I: Sliding Window Training Sample Generation

Input Cycles	Prediction Target
Cycles 1–10	Cycle 11
Cycles 2–11	Cycle 12
Cycles 3–12	Cycle 13
$\vdots$	$\vdots$
Cycles $(N-10)$ – $(N-1)$	Cycle (N)

The window continues to slide until the final valid sequence of n consecutive gait cycles is reached.



Fig. 2: CNN-BiLSTM model architecture

E. Train/Validation/Test Data Splitting

After batch generation, the dataset is split at the file level. This follows a standard 70/15/15 rule, where the procedure is as follows:

- 1) Randomly shuffle the batches within each file.
- 2) Split the data per file into:
 - Training set (70%) – used for model learning
 - Validation set (15%) – used for hyperparameter tuning and monitoring overfitting during training
 - Test set (15%) – used for final performance evaluation
- 3) Combine the corresponding sets across all participant files.

This approach prevents data bias while preserving participant variability across the dataset.

F. Neural Network Design

1) Model Architecture Overview: The model takes the previous $n = 10$ gait cycles as input. Since each gait cycle is resampled to 200 time points and contains 35 feature channels, the input shape becomes $(200 \cdot n, 35, 1)$.

The model predicts the next full gait cycle with output shape $(200, 5)$, corresponding to the three hip joint angles and the PCA-reduced knee and ankle components.

The architecture begins with a series of Conv2D layers that extract local temporal features from the multi-sensor input using progressively decreasing kernel sizes: $(7,1) \rightarrow (5,1) \rightarrow (3,1)$. The feature maps are then reshaped and passed through a dense layer to reduce feature dimensionality, followed by average pooling to reduce the temporal resolution. A BiLSTM layer is subsequently applied to capture long-range temporal dependencies across the entire gait sequence. Finally, a dense layer with linear activation generates the regression outputs, which are reshaped to form the predicted joint angle trajectories.

Figure 2 represents the pipeline for model architecture.

G. Model Training

The model was trained using the following configuration:

- Optimizer: Adam (learning rate = 1×10^{-3})
- Loss function: Mean Squared Error (MSE)
- Evaluation metrics:
 - Mean Absolute Error (MAE) for training
 - Root Mean Squared Error (RMSE) for final performance reporting
- Maximum epochs: 200
- Batch size: 16
- Callbacks:

- ModelCheckpoint (best validation loss)
- EarlyStopping
- ReduceLROnPlateau

III. RESULTS

Five independent trials were run, with a mean test RMSE value of 0.0451 ± 0.002 rad (2.58°). The best run had an RMSE of 0.0437, and the worst run had an RMSE of 0.0482 rad. The mean training time was 1010.35 s (17 min). Table II shows an overall model performance, and Table III shows the RMSE for each of the 5 joint angles.

TABLE II: Overall Model Performance Across 5 Runs

Run	Val_RMSE (rad)	Test_RMSE (rad)	Train Time (s)
1	0.039	0.044	974.5
2	0.038	0.045	905.7
3	0.039	0.045	1054.0
4	0.039	0.048	1060.9
5	0.039	0.044	1056.5
Mean $\pm$ Std	0.039 $\pm$ 0.001	0.045 $\pm$ 0.002	1010.3 $\pm$ 68.6

IV. DISCUSSION

A. Interpretation of Performance

The proposed model demonstrated stable and consistent predictive performance, achieving a mean test RMSE of 0.045 ± 0.002 rad (approximately 2.58°) across five independent runs. This is quite promising, as an average error of less than 3° in a full-range human walking kinematics is considered very accurate.

As shown in Table III, the knee PCA component exhibited the highest prediction error with a mean RMSE of 0.064 rad (approximately 3.67°), while the hip Y component achieved the lowest RMSE score of 0.018 rad (approximately 1.03°). The relatively high error observed in the knee component may be affected by two primary factors.

First, the knee undergoes a larger dynamic range of motion and sharp flexion-extension transitions during the gait cycle, compared to other parts. These larger and more rapid changes can increase sensitivity to small misalignment in prediction, resulting in larger regression errors.

Second, the PCA reduction compresses the original three-axis knee motion into a single dominant component. Although the knee is primarily a hinge joint, real-world gait involves small multi-axis motions and sensor noise that are not perfectly aligned with a single rotation axis. Reducing the motion to one principal component could therefore remove secondary variations, which could contribute to an increased RMSE value.

B. Limitations and Future Work

Despite the precise and low RMSE score results, several limitations are acknowledged for this project. First, the dataset comprises only 7 healthy participants in their 20s or 30s. This limited sample size and narrow demographic range may restrict the ability to generalize the model to broader, larger populations, particularly older people or individuals with motor impairments. Variability could be shown between

the subjects in gait biomechanics, which could impact model performance when applied to unseen participants.

V. CONCLUSION

In the Fall 2025 term, the main objective was to design and train a neural network model capable of predicting lower-limb joint angles for one future gait cycle. The pipeline included reshaping through normalization, PCA-based dimensionality reduction for knee and ankle joints, and sliding window sequence generation, followed by the design and training of a hybrid CNN-BiLSTM neural network.

Across five independent training runs, the proposed model achieved a mean test RMSE of 0.045 ± 0.002 rad (approximately 2.58°), indicating precise and consistent predictive performance. Joint-specific errors remained within a narrow range, with the highest mean RMSE observed for the knee PCA component (0.064 rad) and the lowest for the hip Y component (0.018 rad). The small standard deviation across runs indicates robust convergence and reliable generalization within the dataset.

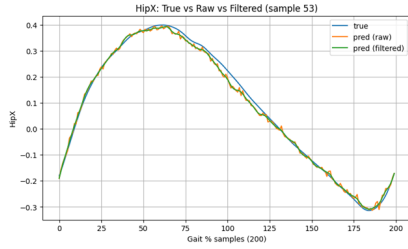
Overall, the results demonstrate that hybrid deep learning architectures leveraging wearable sensor signals can accurately forecast lower-limb kinematics, providing a strong foundation for future real-time gait prediction systems in rehabilitation and assistive robotic applications.

REFERENCES

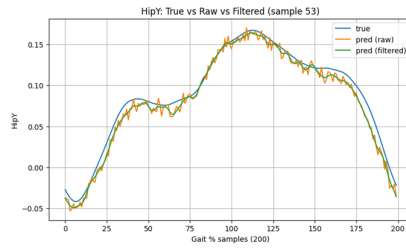
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TABLE III: Test RMSE per Output Channel

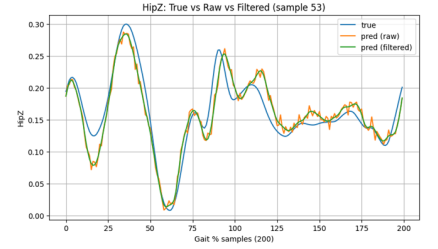
Run	HipX	HipY	HipZ	KneePCA	AnklePCA
1	0.043	0.018	0.041	0.063	0.043
2	0.046	0.018	0.041	0.063	0.044
3	0.046	0.019	0.042	0.064	0.044
4	0.052	0.019	0.042	0.069	0.046
5	0.043	0.018	0.041	0.061	0.044
Mean $\pm$ Std	0.046 $\pm$ 0.004	0.018 $\pm$ 0.001	0.041 $\pm$ 0.000	0.064 $\pm$ 0.003	0.044 $\pm$ 0.001



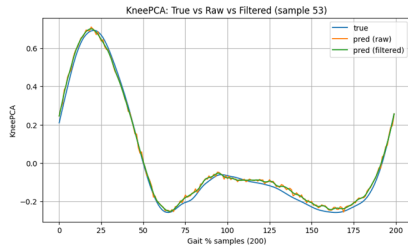
(a) Hip X trajectory



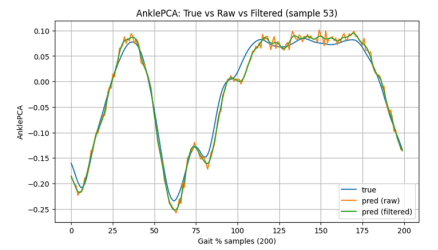
(b) Hip Y trajectory



(c) Hip Z trajectory



(d) Knee PCA trajectory



(e) Ankle PCA trajectory

Fig. 3: Predicted versus ground truth joint angle trajectories for all output channels.